

Database for Orthogonal matrices

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1 Abstract

The subject of this talk has been addressed at this conference in previous years. There are two new developments in the last year. Both are connected to setting up a database of matrices.

The idea which has been constant over the last few years has been the advantages of using rational orthogonal matrices in undergraduate teaching. One point of difficulty for potential users is the question of where do these matrices come from. Early papers described methods whereby an instructor could generate such matrices. Recently, the focus has shifted to creating a public database of such matrices which a user can simply look up. Here we discuss some of the challenges of creating a useful database.

Two particular points arise. First, the question of reducing duplication. It is all too easy to generate a database with redundant entries. Thus the matrices

$$\begin{bmatrix} 1 & 2 & 2 \\ 2 & -2 & 1 \\ 2 & 1 & -2 \end{bmatrix}, \begin{bmatrix} -2 & -1 & 2 \\ 2 & -2 & 1 \\ -1 & -2 & -2 \end{bmatrix}$$

are really equivalent matrices, in that one can be obtained trivially from the other by swapping columns and multiplying some columns by -1 . As the matrices become larger, this equivalence becomes more difficult to identify. We developed a method for doing so.

The second point concerns the density of the matrices. The obvious way to generate matrices leads to dense matrices in which every element is non-zero. We have looked at generating semi-dense matrices, in which a few entries are zero. This gives an instructor a way of controlling the difficulty level of an exercise or question.

Keywords

Orthogonal matrices, Rational matrices, Databases, undergraduate education